

1. How does virtual memory improve CPU utilization and throughput?

1 point

- ☐ By eliminating the need for multiprogramming
- ☐ By loading entire programs into the main memory
- ☐ By increasing the amount of physical memory available
- ☒ By allowing more programs to be loaded into the main memory

2. What is a significant difference between paging and virtual memory?

1 point

- ☐ Paging divides memory into fixed-size blocks, while virtual memory does not.
- ☒ Paging requires the entire program to be in main memory, while virtual memory does not.
- ☐ Virtual memory eliminates the need for physical memory.
- ☐ Paging uses logical addresses, while virtual memory uses physical addresses.

3. What hardware support is necessary to implement demand paging?

1 point

- ☐ Primary memory
- ☐ Frame allocation algorithm
- ☒ Valid-invalid bit
- ☐ High-speed processor

4. Which page replacement algorithm assumes that pages used recently will be needed again soon?

1 point

- ☒ LRU
- ☐ LFU
- ☐ OPT
- ☐ FIFO

5. What is a significant drawback of the Optimal Page Replacement (OPT) algorithm?

1 point

- ☐ It assumes pages used recently will not be needed again
- ☒ It requires future knowledge of memory access patterns
- ☐ It replaces the oldest page in memory
- ☐ It is too simple and often yields poor performance